Asymptotic-center weak convergence is generated by a topology

Literature answer to the open question in arXiv:1406.3215

11 August 2026

Status: literature-already-answered, full positive answer for sequences. Every CAT(0) space has a unique sequential topology $\mathcal{T}_\Delta$ whose convergent sequences are exactly the bounded sequences converging weakly in the asymptotic-center (or Δ-) sense.

Source problem

For a bounded sequence (x_n) in a CAT(0) space X, define

$$\omega(x; (x_n)) = \limsup_{n\to\infty} d(x, x_n).$$

Its unique minimizer is the asymptotic center. The source says that (x_n) converges weakly sequentially to x when x is the asymptotic center of every subsequence. On source PDF page 12, immediately before giving this definition on page 13, Kell asks whether this convergence is actually generated by a topology, citing Bačák's Question 3.1.8 [1].

Explicit later solution

Lytchak–Petrunin explicitly identify the same Kirk–Panyanak/Bačák/Kell question in their introduction. Their Theorem 1.2 states [2]:

For every CAT(0) space X there exists a unique sequential topology $\mathcal{T}_\Delta$ such that a sequence converges to x in $\mathcal{T}_\Delta$ if and only if it is bounded and converges weakly to x.

Their projection definition of weak convergence is equivalent in CAT(0) spaces to the source's requirement that every subsequence have asymptotic center x. Thus Theorem 1.2 is a full affirmative answer to the cited sequential question.

Proof intuition

Declare $A \subset X$ closed when it contains the weak limit of every bounded weakly convergent sequence in A. Arbitrary intersections are immediate. Finite unions work because a sequence in a finite union has a subsequence contained in one member, and weak convergence passes to subsequences. These sets therefore define a topology $\mathcal{T}_\Delta$.

If (x_n) is bounded and weakly converges to x, any subsequence avoiding a neighborhood of x would lie in a sequentially closed complement and force x into that complement. Conversely, a $\mathcal{T}_\Delta$-convergent sequence must be bounded: an unbounded subsequence can be chosen to escape to infinity and its countable range is $\mathcal{T}_\Delta$-closed. If a bounded sequence fails to converge weakly to x, CAT(0) weak compactness supplies a subsequence weakly converging to some $y \neq x$; its range together with y is closed and again contradicts topological convergence to x. The definition also makes $\mathcal{T}_\Delta$ sequential, and convergence of sequences determines any sequential topology uniquely.

Scope boundary

The result concerns sequences, exactly as in the source definition. It does not say that weak convergence of all nets is always induced by a topology. Lytchak–Petrunin show that the net statement requires $\mathcal{T}_\Delta$ to be Hausdorff on every closed ball, and give a bounded CAT(0) complex for which $\mathcal{T}_\Delta$ is not Hausdorff. Non-Hausdorffness does not contradict the positive sequential answer: $\mathcal{T}_\Delta$ is always sequentially Hausdorff.

References

[1] M. Kell, *Uniformly convex metric spaces*, arXiv:1406.3215v1 (2014), pp. 12–13.

[2] A. Lytchak and A. Petrunin, *Weak topology on CAT(0) spaces*, arXiv:2107.09295v2 (2022), Theorem 1.2.